Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/12/2021
PI Name:	Wang, Kegang Ph.D.		
Project Title:	Testing Analytical and Numerical Models in Phase Coarsening using NASA Physical Sciences Informatics System		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32901-6982	**Congressional District:**	8
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	09/16/2016	**End Date:**	09/15/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/15/2020 per NSSC information (Ed., 6/18/20) NOTE: End date changed to 9/15/2019 per NSSC information (Ed., 6/17/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Glicksman, Martin Ph.D. (Florida Institute of Technology)		
Grant/Contract No.:	NNX16AT74G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A quantitative understanding of microstructure evolution by phase coarsening is crucial to the optimization of processing procedures for controlling the final structure and properties of multiphase materials. The volume fraction of the coarsening phase plays an integral role in the kinetics. The existing theoretical work has been limited to some volume fractions. The most ground-based experimental study of phase coarsening has been limited in model alloy systems. There were four microgravity experimental works for phase coarsening in Sn-Pb mixtures. The microgravity experiments greatly reduced the rate of sedimentation and allowed the dynamics of the phase coarsening to be investigated carefully and accurately. The specific thrusts of this project therefore lie in advancing the state of the art of understanding phase coarsening using NASA Physical Sciences Informatics (PSI) System. The proposed research project is to test existing analytical and numerical models for phase coarsening using data in NASA Physical Sciences Informatics System. The primary goals of this project are: (1) We will test the diffusion screening theory of phase coarsening by virtue of the microgravity experiments in NASA Physical Sciences Informatics System; (2) We will test numerical modeling such as phase-field simulation and multiparticle diffusion simulation; (3) Large-scale two-dimensional phase-field simulations will be conducted utilizing the microgravity experiments in NASA Physical Sciences Informatics System; (4) We will carry out a three-way comparison among theory, simulation, and microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is potentially improving our understanding of fundamental materials phenomena, and advancing the state of the art in both theory and experiment as it pertains to microstructural evolution in materials science. Our research will help engineers improve process designs in economically important areas, including liquid-phase sintering for the production of hard tool materials, and industrial crystallization of commodity chemicals and fertilizers, where controlling the particle size distribution underlies important safety and manufacturing issues. The improved knowledge of phase coarsening in this project will have significant impact on alloy design and service-life strategies for high-temperature materials. Our work has affected multiple disciplines such as materials science, chemistry, and physics within the field of statistical mechanics modeling for pattern formation and evolution. This simulation method and theory are of significant interest for the physical sciences and technologies community.
Task Progress:	We have successfully developed a 2-D phase-field simulation code and a multi-particle diffusion simulation code for phase coarsening simulations, and a Matlab code for image analysis and extraction of data. Using the Matlab code, we extracted concentration and orientation distributions from the initial microstructures measured at the earlier time archived from the Coarsening in Solid-Liquid Mixtures (CSLM) experiments. To conduct quantitative comparison of simulations with archived experimental data from CSLM, NASA data were used as input for our phase-field code, allowing simulated evolution of coarsened microstructures for the Sn-Pb alloy mixtures processed aboard International Space Station (ISS). Using our multi-particle diffusion simulation code, we have accomplished 3-D multiparticle diffusion simulations for phase coarsening at a volume fraction of 0.15. We compared these simulation results qualitatively with archived data derived from CSLM experiments, and theoretical predictions from our diffusion screening theory. Using our 2-D phase-field simulation code, we have conducted 2-D phase-field simulations for phase coarsening at volume fractions of 0.15, 0.3, 0.7, 0.8. We compared these simulation results with archived data derived from CSLM experiments. We compared the evolved microstructures from phase-field simulations with those recorded from the CSLM experiments for those volume fractions archived in NASA's PSI system. In order to check the accuracy of this advanced theory of phase coarsening, we measured values of the scaled maximum particle sizes by average size, Rmax. For volume fraction of 0.15, our simulation and diffusion screening theory, and CSLM experiment predicted that Rmax=1.63, 1.68, 1.58, respectively. For volume of 0.3, simulation and diffusion screening theory predicted that Rmax=1.77, and 1.73, respectively, and CSLM experiment yielded Rmax=1.84. These values are in good agreement given both the uncertainties in the statistical experimental and theoretical estimates. From our microstructures generated from phase-field simulation, the particle size distribution (PSD) can be calculated. From microgravity experimental microstructures the PSD was calculated by CSLM group. In addition, our diffusion screening theory can predict the PSD for volume fraction of 0.15. We compared the PSDs among simulations, theories, and microgravity experiments at volume fraction of 0.15, which is in a good agreement with that from microgravity experiments, and in a reasonable agreement with the prediction of the diffusion screening theory.
Bibliography Type:	Description: (Last Updated: 10/30/2021)
Articles in Peer-reviewed Journals	Wang KG, Wang GQ, Gamsjager E, Glicksman ME. "Quantitative comparison of theory and simulation with microgravity experiments in phase coarsening. A comparison of theory and simulation with microgravity experiments on phase coarsening." Acta Materialia.2021 Dec;221:117402. https://doi.org/10.1016/j.actamat.2021.117402 [NOTE previously listed in Jan 2021 as "Submitted as of December 2020.'] , Dec-2021